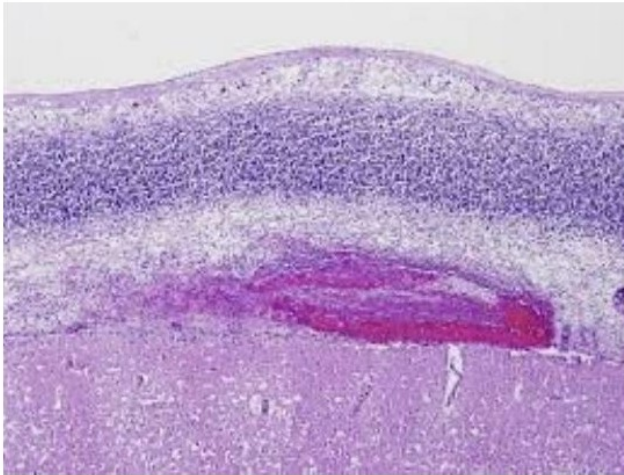


Lecture 1 — Companion Figures

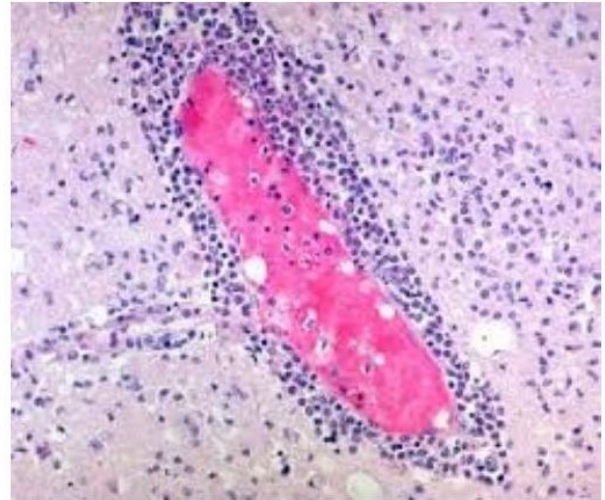
Cerebral Inflammation

Prof Steve Gentleman — Imperial

Meningitis



Viral encephalitis



neuropathology-web.org

Figure L1.1 — Meningitis vs viral encephalitis histology: subarachnoid immune-cell exudate vs perivascular cuffing.

the existence of a Blood-Brain Barrier.



Organs removed from a mouse after intravenous injection with a blue dye.

Dye did not accumulate in brain

Figure L1.2 — Historical demonstration of the BBB: IV dye stains all visceral organs but spares the brain.

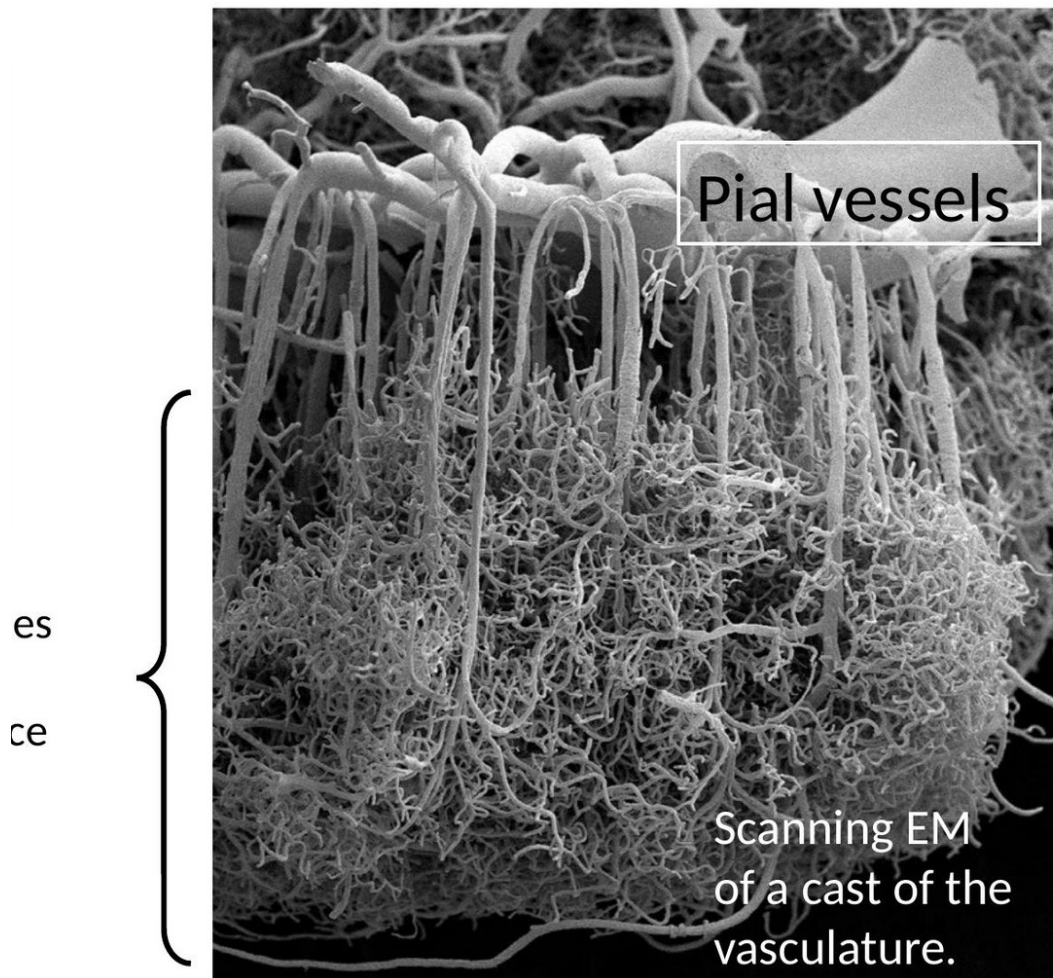


Figure L1.3 — Scanning EM cast of CNS vasculature: pial vessels send capillary branches deep into brain substance.

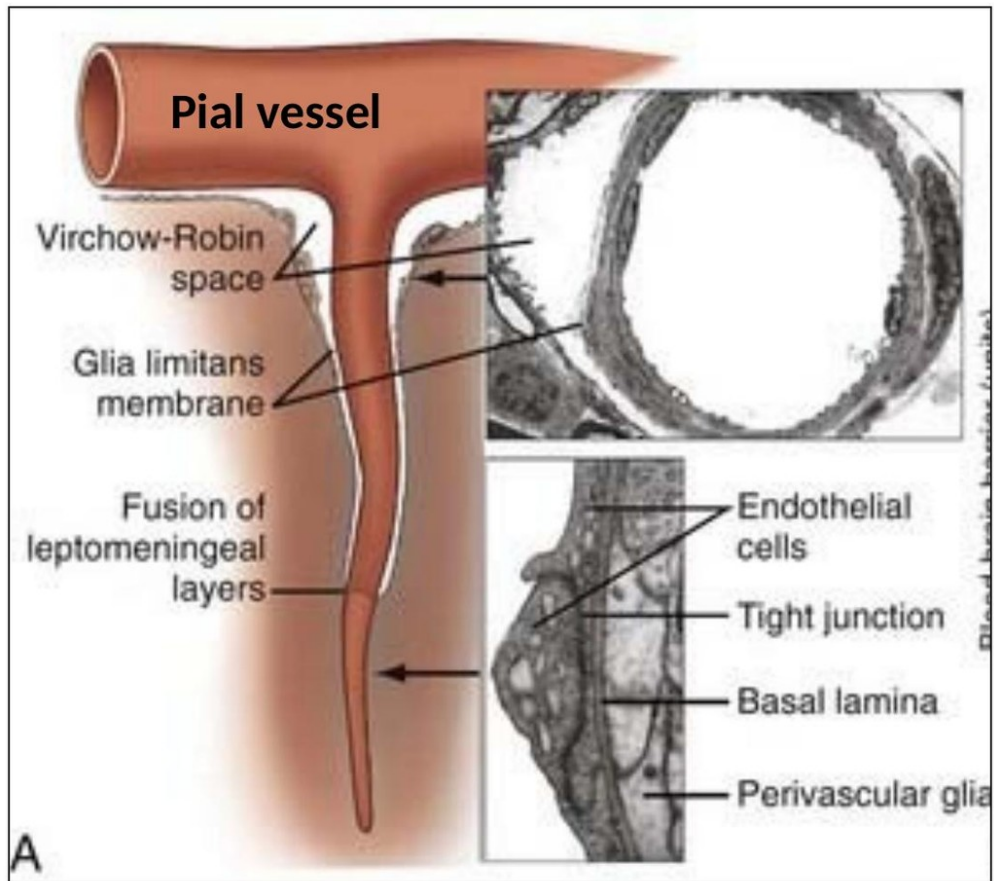


Figure L1.4 — Anatomy of the BBB: pial vessel, Virchow-Robin space, glia limitans, endothelial tight junctions, basal lamina.

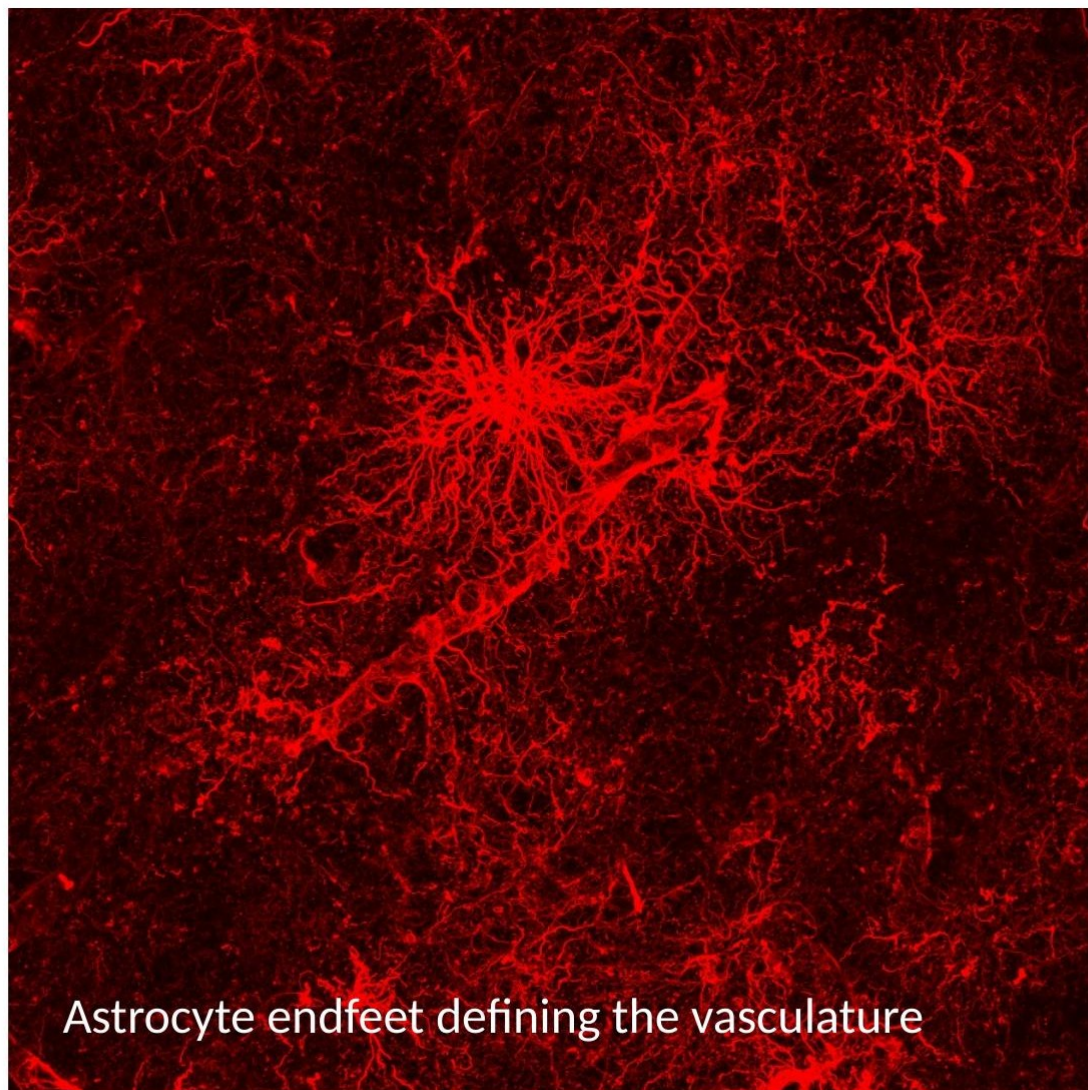


Figure L1.5 — Astrocyte end-feet defining the cerebral vasculature (GFAP fluorescence). The vessel itself is unstained — the end-feet outline it.

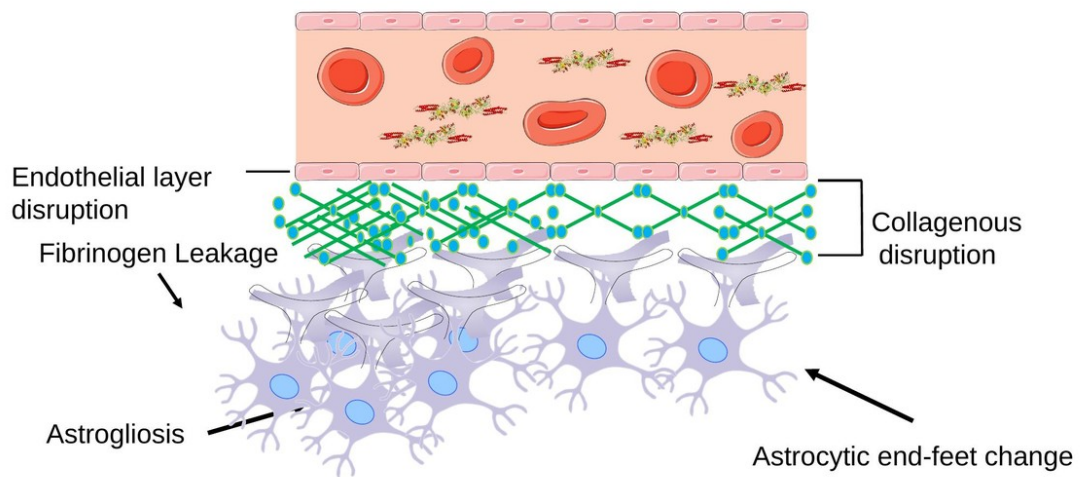
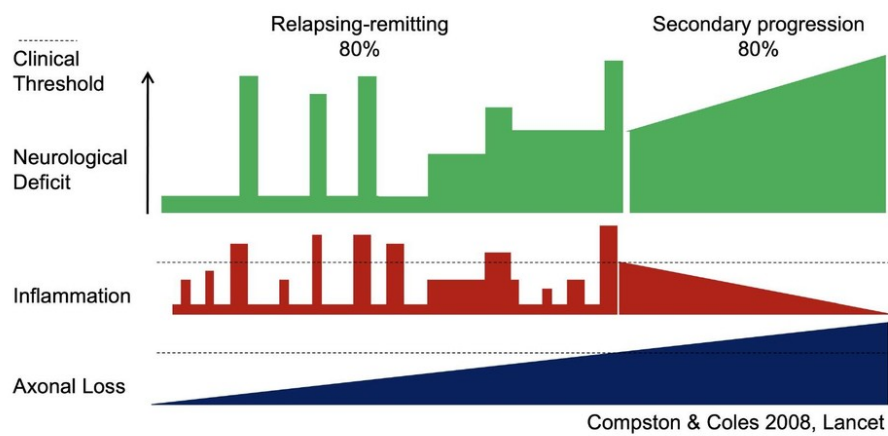


Figure L1.6 — BBB disruption sequence: endothelial breakdown to fibrinogen leak to astrocyte end-feet retraction to collagenous disruption + astrogliosis.



- Autoimmune demyelinating disease of the CNS

Figure L1.7 — MS disease course (Compston & Coles 2008): relapsing-remitting then secondary progression, with overlaid inflammation peaks and steady axonal loss.

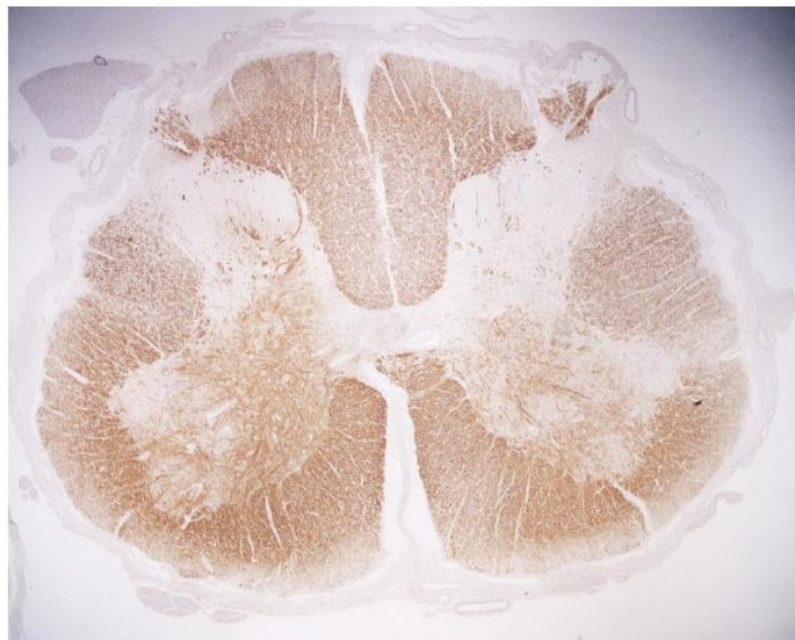
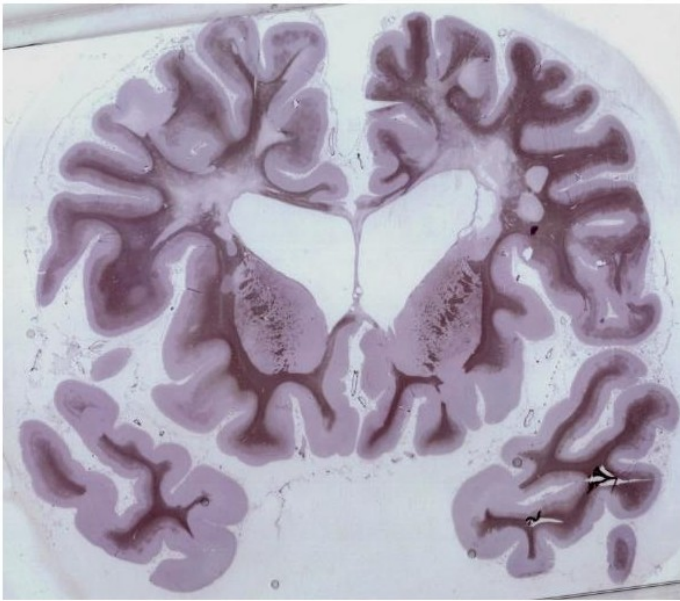


Figure L1.8 — MS pathology: periventricular plaques on coronal myelin stain (top); demyelinated patches in spinal cord white matter (bottom).

RI Scans of the Brain of a 25-Year-Old Woman with Relapsing–Remitting Multiple Sclerosis

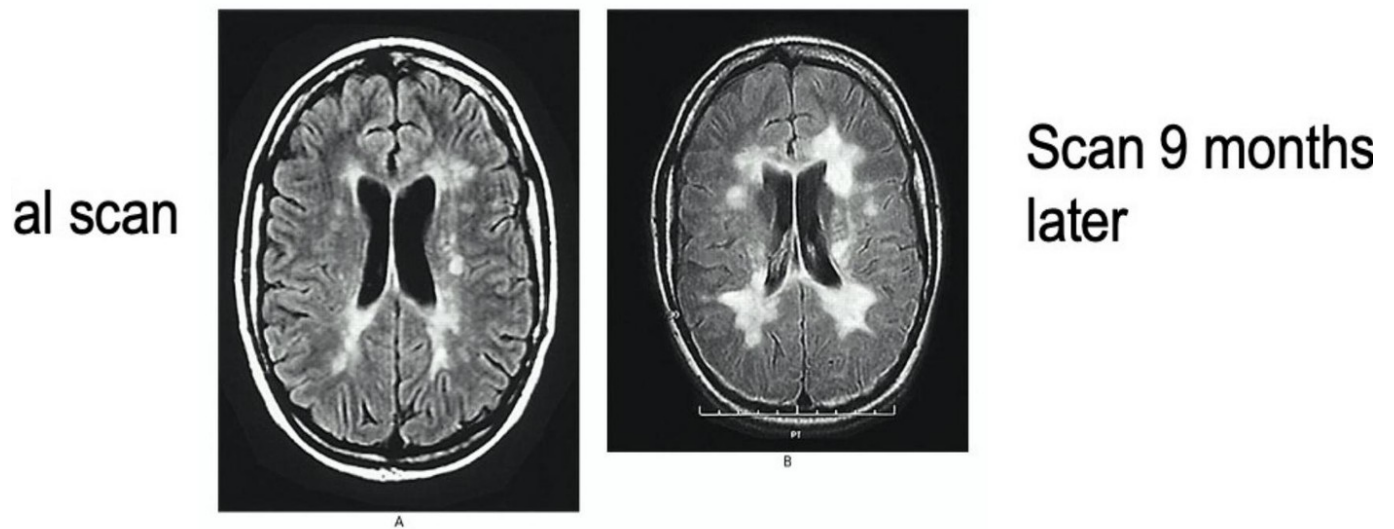


Figure L1.9 — MRI of relapsing-remitting MS: periventricular white-matter lesions enlarge over 9 months — dissemination in time.

Symptoms vary because the amount and location of damage to the nervous system is different in each person with MS

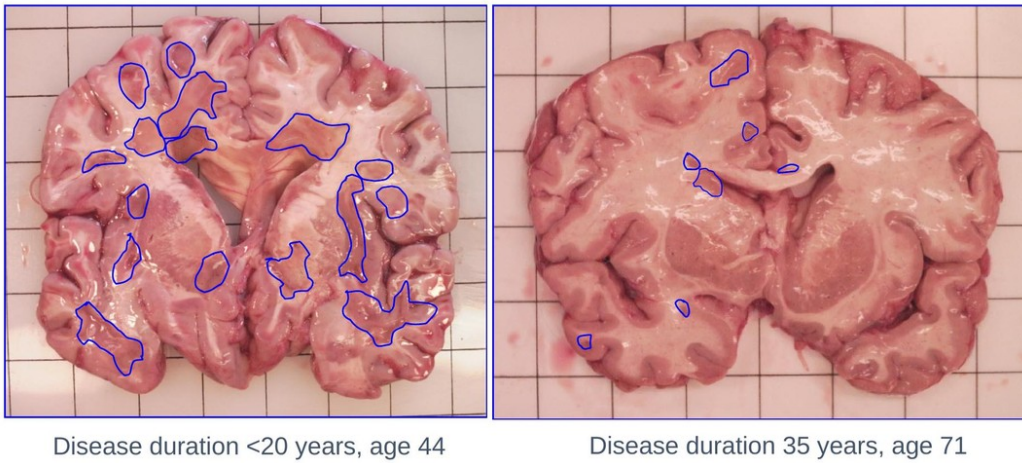
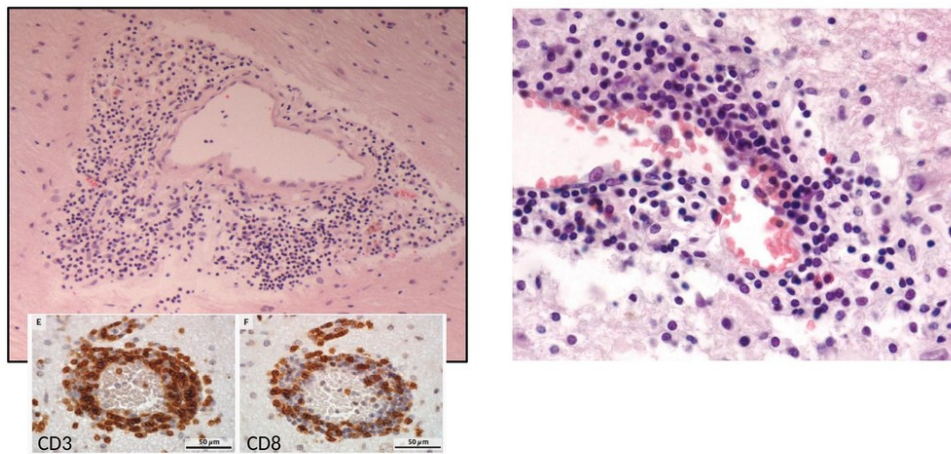


Figure L1.10 — Coronal brain sections from two MS patients: distribution and quantity of plaques varies with disease duration and individual.



-perivascular immune cell infiltration (CD3 T-cells and CD20 B-cells)

Figure L1.11 — Perivascular immune-cell cuffing in MS: T-cell (CD3, CD8) and B-cell (CD20) infiltration around a vessel.